TruLearnix

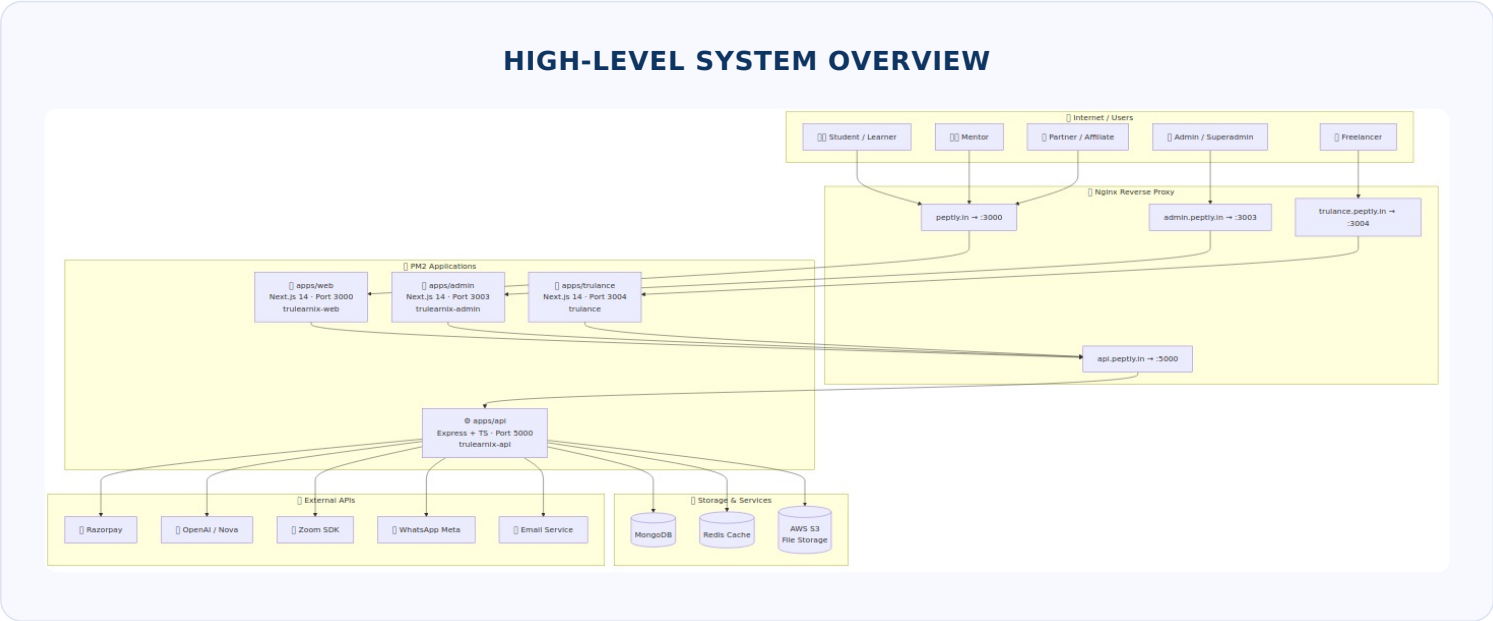
Platform Architecture Document

peptly.in · admin.peptly.in · api.peptly.in · trulance.peptly.in

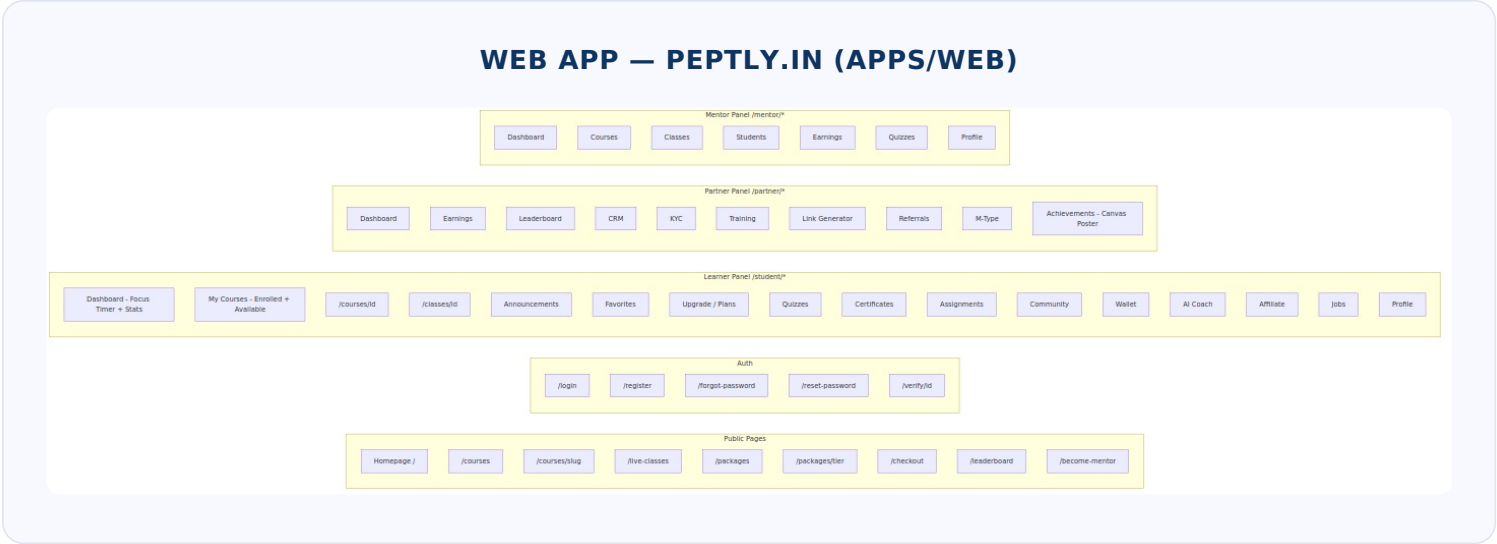
TruLearnix — Platform Architecture

> *Ed-tech + Freelance + Affiliate platform running at **peptly.in***

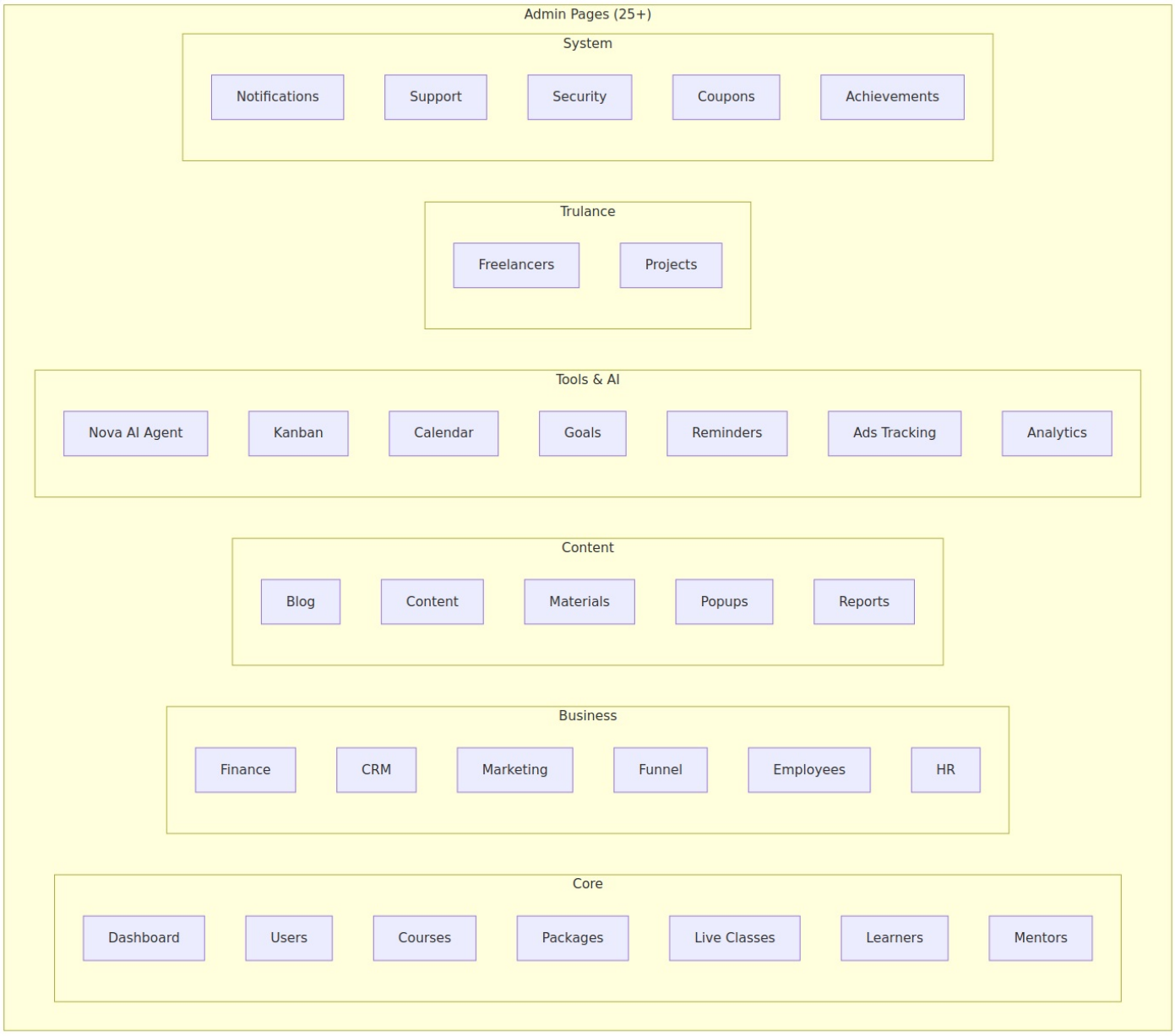
High-Level System Overview



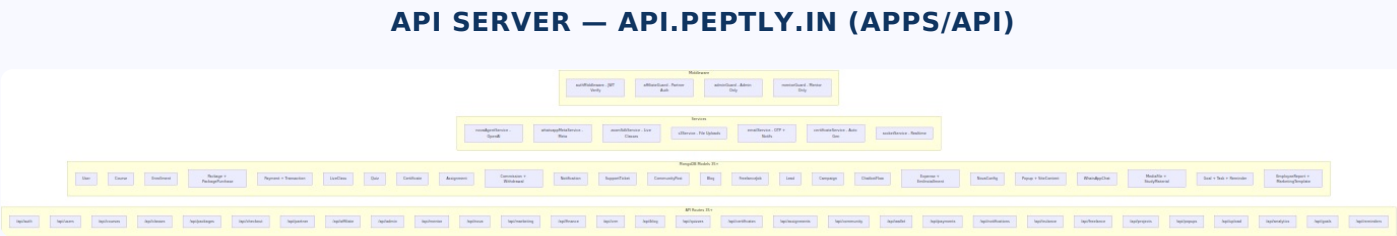
Web App — peptly.in (`apps/web`)



Admin Panel — admin.peptly.in (`apps/admin`)

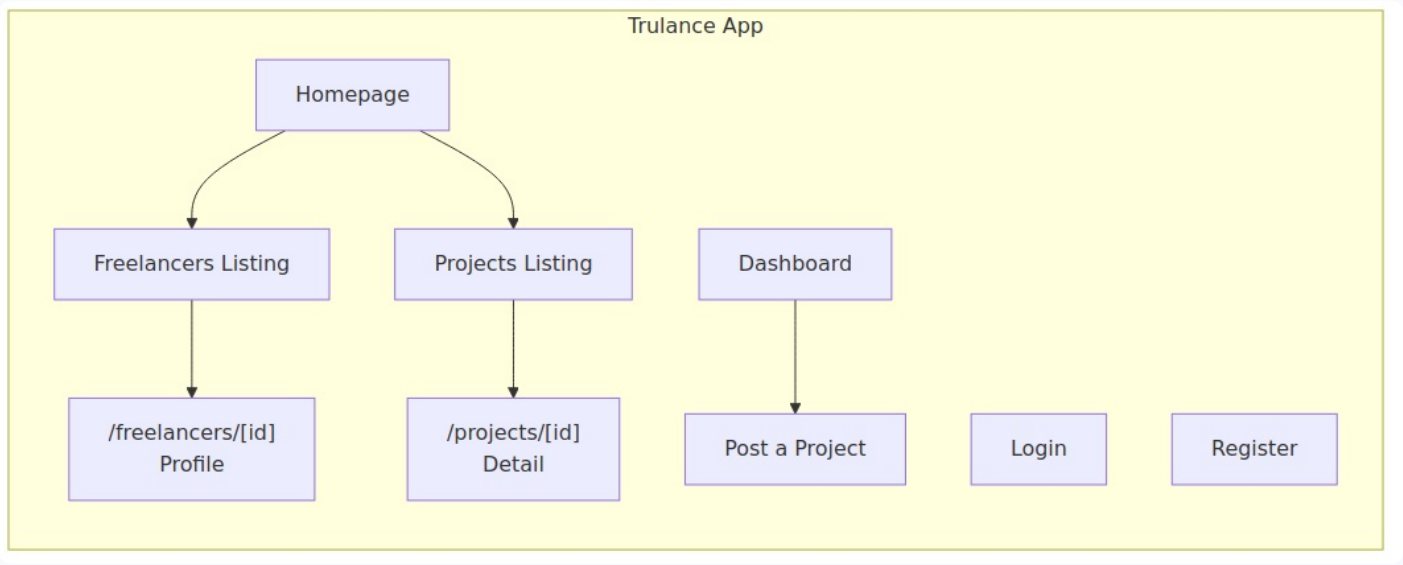


⚙ API Server — api.peptly.in (`apps/api`)



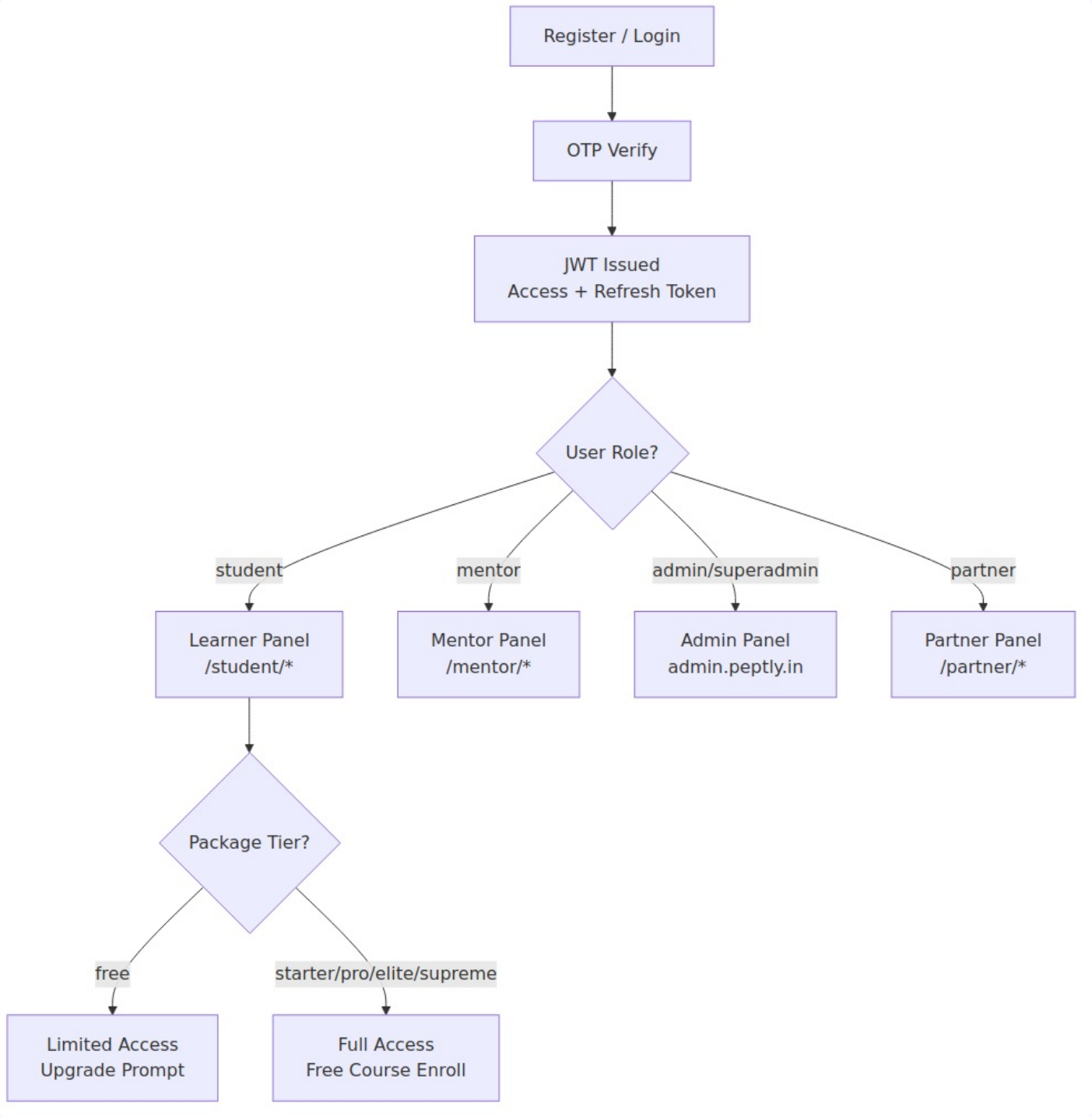
Trulance — Freelance Marketplace (`apps/trulance`)

TRULANCE — FREELANCE MARKETPLACE (APPS/TRULANCE)

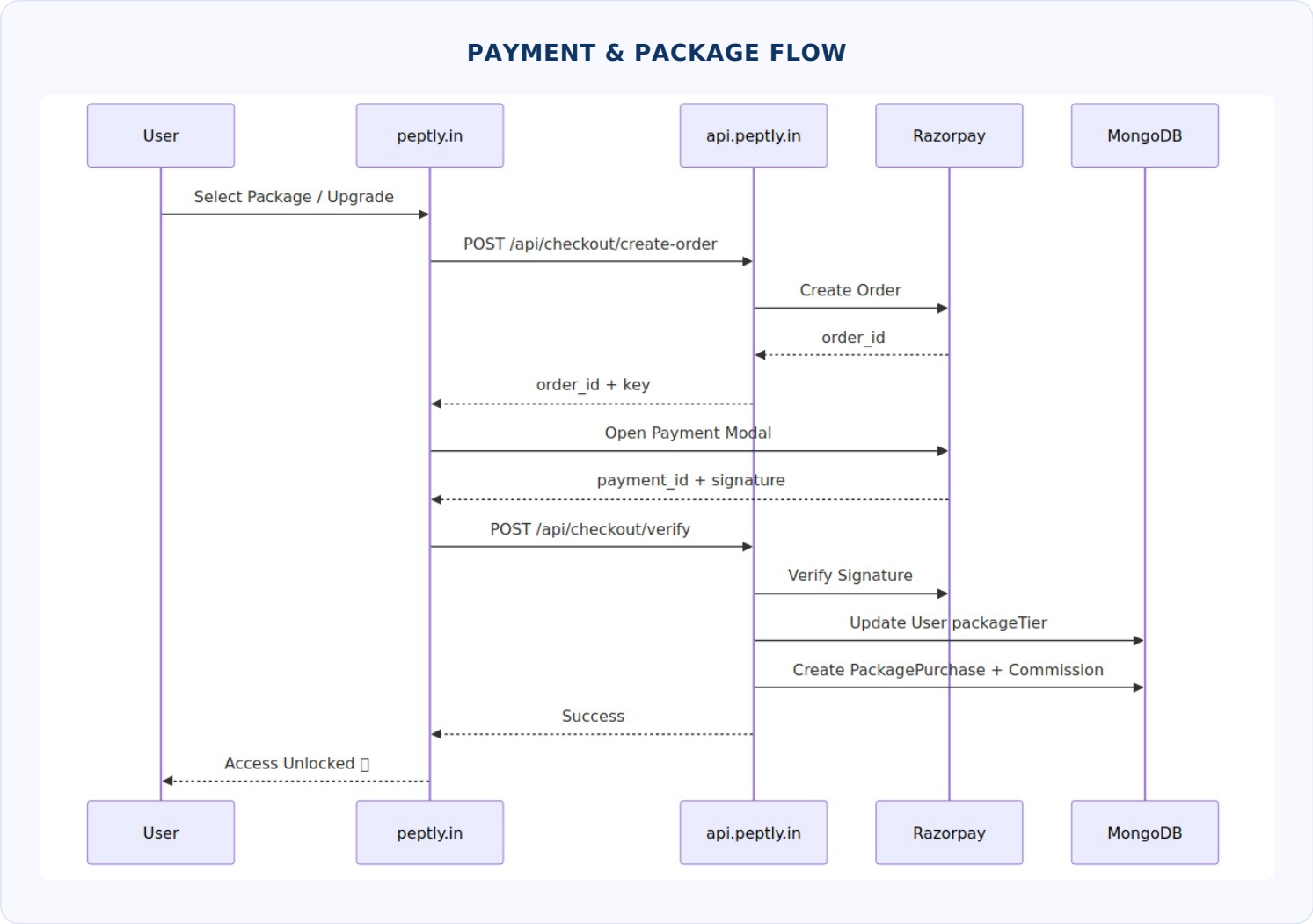


Auth & User Role Flow

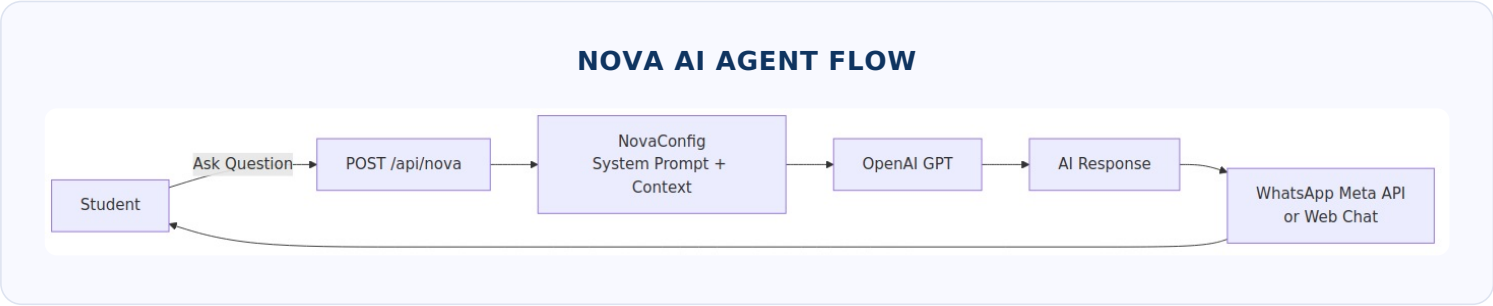
AUTH & USER ROLE FLOW



Payment & Package Flow



Nova AI Agent Flow



Monorepo Directory Structure

```
trulearnix/
├── apps/
│   ├── web/                    # peptly.in — Next.js 14 (Port 3000)
│   │   ├── app/
│   │   │   ├── (auth)/         # login, register, forgot-password
│   │   │   ├── (public)/       # homepage, courses, packages, checkout
│   │   │   ├── (student)/      # /student/* — Learner Panel (20 pages)
│   │   │   ├── (mentor)/       # /mentor/* — Mentor Panel (7 pages)
```

```
│   │   │   ├── (partner)/      # /partner/* — Partner Panel (11 pages)
│   │   │   └── become-mentor/
│   │   ├── components/
│   │   │   ├── layout/         # Navbar, Footer, LearnerSidebar, PartnerSidebar
│   │   │   ├── popups/         # PopupManager, AnnouncementPopup, EarningsToast
│   │   │   ├── shared/         # HeroSection, Stats, Testimonials, Pricing...
│   │   │   └── ui/             # Logo, Tilt3D, buttons...
│   │   └── lib/                # api.ts, store.ts (Zustand), affiliateData.ts
│   │
│   ├── admin/                  # admin.peptly.in — Next.js 14 (Port 3003)
│   │   ├── app/                # 25+ admin pages
│   │   └── components/         # AdminLayout, Sidebar
│   │
│   ├── api/                    # api.peptly.in — Express + TypeScript (Port 5000)
│   │   └── src/
│   │       ├── models/         # 35+ MongoDB models
│   │       ├── routes/         # 35+ API route files
│   │       ├── controllers/    # Business logic
│   │       ├── services/       # Nova, WhatsApp, Zoom, S3, Email, Socket
│   │       └── middleware/     # auth, affiliateGuard, adminGuard
│   │
│   └── trulance/               # Freelance Marketplace — Next.js 14 (Port 3004)
│       ├── app/                # 8 pages
│       └── components/         # Navbar
│
├── nginx/                      # Nginx config files
├── ecosystem.config.js         # PM2 process manager config
└── package.json                # Monorepo root
```

Tech Stack Summary

Layer	Technology

|-------|-----------|

Frontend	Next.js 14, TypeScript, TailwindCSS
State Management	**Zustand, React Query 5**
Backend	Express.js, TypeScript
Database	MongoDB (local)
Cache	Redis (local)
File Storage	AWS S3
Auth	JWT (Access + Refresh Token), HTTP-only Cookies
Payments	Razorpay
AI	OpenAI (Nova Agent)
Live Classes	Zoom SDK + WebRTC
Messaging	WhatsApp Meta Cloud API
Real-time	Socket.io
Process Manager	PM2

Reverse Proxy	Nginx
Deployment	VPS (Hostinger)